Review: Multivariate time series clustering based on common principal component analysis[1]

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1 Introduction and contributions

This paper introduces a novel method for clustering multivariate time series (MTS), inspired by the K-means algorithm and grounded in Common Principal Component Analysis (CPCA). The proposed approach follows an iterative two-step procedure: (1) for each cluster, a cluster-specific projection space is constructed by estimating common principal components from the covariance matrices of the MTS assigned to that cluster; (2) each MTS is then assigned to the cluster whose projection space yields the smallest reconstruction error.

Clustering is a fundamental task in machine learning, including in the context of time series analysis. However, clustering multivariate time series remains particularly challenging due to their high dimensionality and complex inter-variable dependencies. A common practical strategy consists in treating each univariate time series independently and processing them separately [2]. While this approach can be effective in some cases, it fails to exploit shared dependencies across variables and samples, and often leads to high computational costs as dimensionality increases. This limitation is especially apparent in DTW-based clustering methods [3], where the repeated computation of cluster prototypes significantly impacts scalability.

Principal Component Analysis (PCA) [4] is a widely used technique to address high-dimensional data by providing compact representations. In the context of multivariate time series, where one observes N time series $x_t^{(1)}, \ldots, x_t^{(N)}$ of length T, PCA aims to identify a reduced set of $k < N$ principal components that capture most of the signal variance. The quality of such representations is typically assessed by measuring similarity between time series in the projected space: two similar groups or series should be more similar than other unrelated. In this article, the authors propose an efficient clustering approach that compares similarity not between the projections but between the original MTS and the reconstructed one. The goal is also to emphasize the comparison to DTW-based approaches.

In this report, we first reproduce the results presented in the original paper on two datasets and extend the evaluation to an additional dataset. We then complement the analysis by introducing two additional evaluation metrics, studying the impact of missing values, and assessing robustness with respect additive noise, multiple initializations and runs. Regarding our contributions, we jointly analyzed the paper and designed the experiments. One group member focused on the introduction and the model, while the other concentrated on experiments and results. As no official implementation was provided by the authors, we also reimplemented the method and made our code publicly available at https://github.com/luckyman94/Implementation-of-the-MC2PCA.

2 Method

In this section, we describe the method proposed by the authors. We first explain the model and then the clustering approach.

2.1 Model

Common principal component analysis (CPCA) is a variant of the Principal component analysis (PCA). While PCA finds the principal components of a single MTS, CPCA finds a single set of principal components that are common to multiple MTS, enabling comparison and clustering in a shared coordinate space.

Let $X = \{X_1, \ldots, X_N\}$ denote a dataset of multivariate time series, where each series $X_i \in \mathbb{R}^{n_i \times m}$ consists of m variables observed over n_i time steps. Although the time series may have different lengths, they share the same number of variables. For each time series X_i, a covariance matrix is computed after centering each variable, yielding $\Sigma_i = \mathrm{cov}(X_i) = \mathbb{E}[\bar{X}_i^\top \bar{X}_i]$ where $\bar{X}_i$ denotes the mean-centered version of X_i. The shared low-dimensional representation across multiple MTS is constructed by using a common covariance matrix, defined as the average of individual covariance matrices $\bar{\Sigma} = \frac{1}{N} \sum_{i=1}^{N} \Sigma_i$.

An eigenvalue decomposition (or singular value decomposition) of $\bar{\Sigma}$ yields a set of eigenvectors $U = [U_1, \ldots, U_m]$ associated with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_m$. Retaining the first p eigenvectors produces a common projection matrix $S = [U_1, \ldots, U_p] \in \mathbb{R}^{m \times p}$, which defines a shared coordinate space capturing the dominant variance structure across the dataset.

Once the common projection space is obtained, any multivariate time series X_i can be projected onto this space as $P_i = X_i S$, resulting in a reduced representation $P_i \in \mathbb{R}^{n_i \times p}$. Using the orthonormality of the eigenvectors, the projected data can be mapped back to the original space to form a reconstructed time series $Y_i = P_i S^\top = X_i S S^\top$.

Since dimensionality reduction generally induces information loss, the discrepancy between the original and reconstructed series is quantified by the reconstruction error $E_i = \|X_i - Y_i\|_2^2 = \sum_{j=1}^{n_i} \sum_{k=1}^{m} (y_{jk} - x_{jk})^2$. This reconstruction error depends both on the number of retained components p and on the quality of the common projection space. In the proposed method, it serves as a measure of how well a given time series is represented by a specific common subspace and is later used for clustering assignment.

2.2 Algorithm

The proposed clustering method, referred to as *Mc2PCA*, takes its inspiration from the classical K-means paradigm but replaces the notion of a centroid with a *subspace-based prototype*, i.e the $S_k \in \mathbb{R}^{n_i \times p}$. It performs clustering of multivariate time series by iteratively estimating cluster-specific projection subspaces and assigning series based on reconstruction error.

Let $X = \{X_1, \ldots, X_N\}$ be a dataset of MTS, where $X_i \in \mathbb{R}^{n_i \times m}$, and let K denote the number of clusters.

1. Initialization The dataset is initially partitioned into K clusters, either randomly or by uniform assignment. For each time series X_i, a covariance matrix $\Sigma_i = \mathrm{cov}(X_i)$ is computed once after mean-centering.

2. Estimation of the cluster prototypes At each iteration, and for each cluster C_k, the algorithm computes a cluster-specific prototype in the form of a projection subspace. This is achieved by averaging the covariance matrices of the MTS assigned to the cluster, $\bar{\Sigma}_k = \frac{1}{|C_k|} \sum_{X_i \in C_k} \Sigma_i$ followed by an eigenvalue decomposition of $\bar{\Sigma}_k$. The first p eigenvectors define the projection matrix $S_k \in \mathbb{R}^{m \times p}$, which represents the common principal components of cluster k.

3. Assignment to the clusters Each MTS X_i is evaluated against every cluster prototype S_k. For each cluster, the series is projected and reconstructed as $Y_{ik} = X_i S_k S_k^\top$, and the corresponding reconstruction error is computed: $E_{ik} = \|X_i - Y_{ik}\|_2^2$. The time series X_i is assigned to the cluster that minimizes this error, $k' = \arg\min_{k=1,\dots,K} E_{ik}$.

The estimation and assignment steps are repeated until reaching a number of given iterations or by plateauing at a negligible change in the total reconstruction error between $E^{(t)}$ and $E^{(t-1)}$.

The overall complexity of Mc2PCA is linear in the number of multivariate time series N. Precomputing covariance matrices costs $O(N\,n\,m^2)$, while each iteration requires $O(Km^3)$ to estimate the common projection axes S and $O(NKnmp)$ for assignment. In practice, since $p \ll m \ll N$, the method scales efficiently to large MTS datasets.

3 Data

To evaluate our implementation, we first chose two datasets used in the article to see if we could reproduce the results. We also used the Basic Motions dataset [5], which is a very small multivariate time series dataset describing human motion patterns recorded by multiple sensors. This dataset allows us to analyze the behavior and stability of MC2PCA in a low-sample regime, where covariance estimation may become unreliable.

The **Japanese Vowels** dataset [6] consists of 640 multivariate time series with 12 features extracted from utterances produced by nine male speakers. The features correspond to Linear Prediction Coding (LPC) cepstrum coefficients obtained through frame-wise linear prediction analysis of the speech signal[7]. From a data analysis perspective, these features exhibit strong inter-feature correlations related to vocal tract dynamics, making this dataset particularly suitable for CPCA-based methods. The underlying assumption in this dataset is that speakers produce consistent spectral patterns across utterances, leading to speaker-specific covariance structures.

The **Spoken Arabic Digits** dataset [8] contains 8,800 multivariate time series corresponding to spoken digits from 44 male and 44 female native Arabic speakers. Each time series is represented by 13 mel-frequency cepstral coefficients (MFCCs)[9]. These features are known to be robust to noise and small variations in pronunciation [10]. As a result, this dataset presents a different covariance structure compared to Japanese Vowels, with smoother inter-feature dependencies and greater intra-class variability, making it a complementary benchmark for evaluating MC2PCA.

The **Basic Motions** dataset comprises only 80 samples distributed across four motion classes, each described by six sensor-derived features. Unlike the speech datasets, the time series are of fixed length and originate from physical motion measurements. This lower dimensionality and more limited number of samples per class could make this dataset more challenging for the covariances estimation.

Dataset	Samples	Classes	Features	Domain
Japanese Vowels	640	9	12	Speech
Spoken Arabic Digits	8800	10	13	Speech
Basic Motions	80	4	6	Human Motion

Table 1: Overview of the multivariate time series datasets used in our experiments.

4 Results

4.1 Metrics used

To quantitatively evaluate the clustering performance of MC2PCA, we rely on standard clustering metrics. We report the **Adjusted Rand Index (ARI)** and the **Normalized Mutual Information (NMI)**, which measure the agreement between predicted clusters and ground-truth labels while correcting for chance. These complementary metrics are widely used in the clustering literature: ARI emphasizes pairwise assignment consistency, whereas NMI captures the mutual dependence between partitions.

In addition, following [1], we report cluster-based metrics, namely Precision and Recall. These measures assess the consistency between predicted clusters and true classes and are particularly relevant when the number of clusters is fixed. In [1], Precision is the primary evaluation criterion used to compare MC2PCA with competing methods such as KMeans-DTW and Spectral-DTW.

4.2 Experiments from the paper

In accordance with the experimental setup of the original paper, we fix the number of clusters to the ground-truth number of classes for each dataset. We evaluate the impact of the CPCA subspace dimension p by varying it from 1 to 5, as done in [1]. For each configuration, clustering results are averaged over multiple random initializations to account for the sensitivity of MC2PCA to initialization.

All quantitative results are summarized in Table 2, which reports the clustering performance for the three datasets considered.

For both the Japanese Vowels and Spoken Arabic Digits datasets, our Precision values are slightly lower but very close to those reported in [1]. In particular, Li reports Precision values of about 0.56 for Japanese Vowels and 0.42 for Spoken Arabic Digits, while our best results (obtained for $p = 3$) reach 0.56 and 0.41, respectively. These minor differences are likely due to implementation details and to averaging over multiple random initializations, whereas the original paper reports single-run results. Overall, both the trends and the performance levels are consistent with the original study.

4.3 Other experiments

To further deepen our analysis, we conducted several additional experiments on the three datasets in order to study the robustness and sensitivity of MC2PCA under different perturbations

First, on the **Japanese Vowels** dataset, we investigated the influence of the CPCA subspace dimension by varying the number of retained components from 1 to 10. As shown in Figure 1, the ARI increases for small values of p, reaches a maximum for intermediate dimensions and then decreases when too many components are retained. This behavior suggests that a limited number of common principal components is sufficient to capture the discriminative structure of the data, while larger subspaces may introduce noise and degrade clustering performance.

We then evaluated the robustness of MC2PCA to additive Gaussian noise by progressively corrupting the time series with noise levels ranging from 0% to 50%. As illustrated in Figure 2, the ARI remains relatively stable for low to moderate noise levels, and only degrades significantly for strong perturbations. This indicates that the method is reasonably robust to additive noise, which is consistent with the covariance-based nature of CPCA.

In contrast, when introducing missing values in the data, the clustering performance deteriorates much more rapidly. Figure 3 shows a clear decrease in ARI as the proportion of removed observations increases, highlighting the strong dependence of MC2PCA on reliable covariance estimation and complete temporal information.

Finally, motivated by the speech nature of the dataset, we studied the impact of truncating either the beginning or the end of the signals. These temporal truncations did not lead to a strong or systematic change in ARI, suggesting that the most discriminative information is distributed across the signal rather than being concentrated in a specific temporal region.

On the **Basic Motions** dataset, we assess the stability of MC2PCA under random initializations. For a fixed CPCA dimension ($p = 3$) and a fixed number of clusters equal to the ground-truth classes, the algorithm is run multiple times using different random permutations of the training set.Clustering performance is evaluated using the ARI and the NMI. The results indicate a moderate yet consistent clustering performance, with ARI values concentrated around 0.24 and a limited variability across runs. This suggests that, although the overall separation between classes remains challenging on this dataset, MC2PCA exhibits a reasonable robustness to initialization effects. In addition, the distribution of cluster sizes across different runs shows no evidence of severe cluster collapse, despite the relatively small size of the dataset.

This behavior is further supported by the convergence curves shown in Figure 4, which illustrate a stable and rapid decrease of the reconstruction error across initializations.

Finally, on the *Spoken Arabic Digits* dataset, increasing the number of clusters from $K = 8$ to $K = 11$ leads to a higher ARI (Figure 5), as values closer to the true number of classes better separate the digit categories. This also highlights the sensitivity of MC2PCA to the choice of K.

In summary, the observed trends across datasets and experimental settings provide a coherent picture of the behavior of MC2PCA, confirming its robustness while revealing its sensitivity to structural and temporal degradations of the data.

4.4 Discussion and Limitations

A key limitation of MC2PCA is its reliance on accurate covariance estimation. As shown by our missing-data experiments, clustering performance degrades quickly when observations are removed, which is inherent to CPCA-based methods relying on second-order statistics. This issue can be mitigated using robust covariance estimators, such as shrinkage-based approaches [11].

Another limitation of MC2PCA concerns the choice of hyperparameters, in particular the number of retained components p and the number of clusters K. Although our experiments show that reasonable values lead to stable results, inappropriate choices may noticeably degrade clustering performance.

Finally, MC2PCA assumes that the main discriminative information between time series can be captured through linear correlations encoded in covariance matrices. This assumption may limit its applicability to datasets exhibiting strong non-linear temporal dependencies.

A Experimental results

Dataset	p	ARI	NMI	Precision	Recall
Spoken Arabic Digits	1	0.163	0.285	0.338	0.436
	2	0.172	0.316	0.380	0.389
	3	0.190	0.337	0.414	0.397
	4	0.183	0.323	0.384	0.390
	5	0.144	0.260	0.346	0.346
Basic Motions	1	0.277	0.487	0.575	0.775
	2	0.201	0.289	0.500	0.650
	3	0.182	0.328	0.512	0.575
	4	0.155	0.233	0.500	0.650
	5	0.201	0.296	0.500	0.738
Japan Vowels	1	0.269	0.412	0.537	0.498
	2	0.195	0.325	0.431	0.417
	3	0.318	0.438	0.564	0.559
	4	0.287	0.412	0.519	0.511
	5	0.391	0.484	0.627	0.616

Table 2: Clustering performance of MC2PCA for different values of the CPCA dimension p on the three datasets.

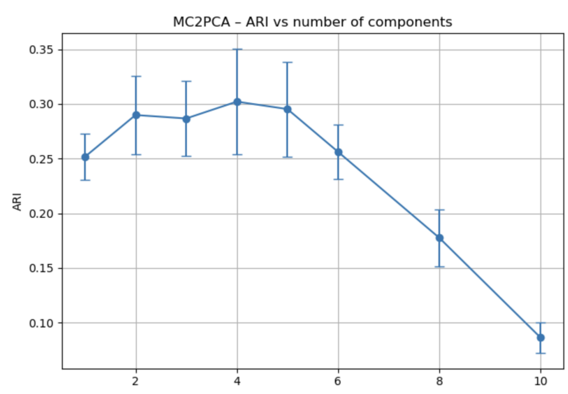

Figure 1: Influence of the CPCA subspace dimension p on clustering performance (ARI) for the Japanese Vowels dataset.

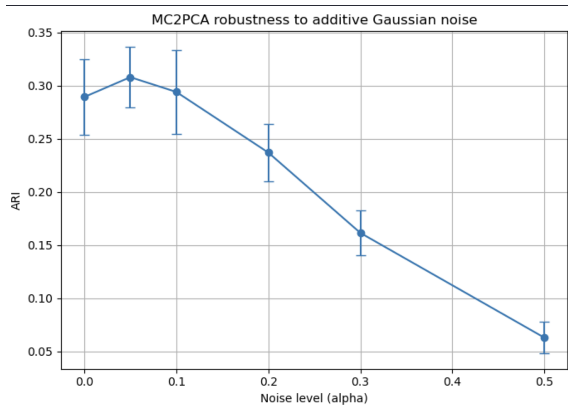

Figure 2: Clustering performance (ARI) under additive Gaussian noise for the Japanese Vowels dataset.

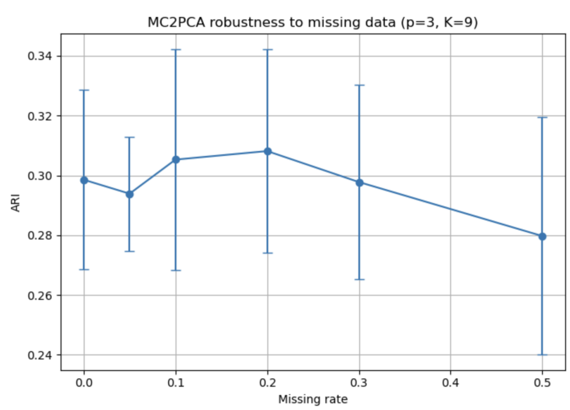

Figure 3: Impact of missing values on clustering performance (ARI) for the Japanese Vowels dataset.

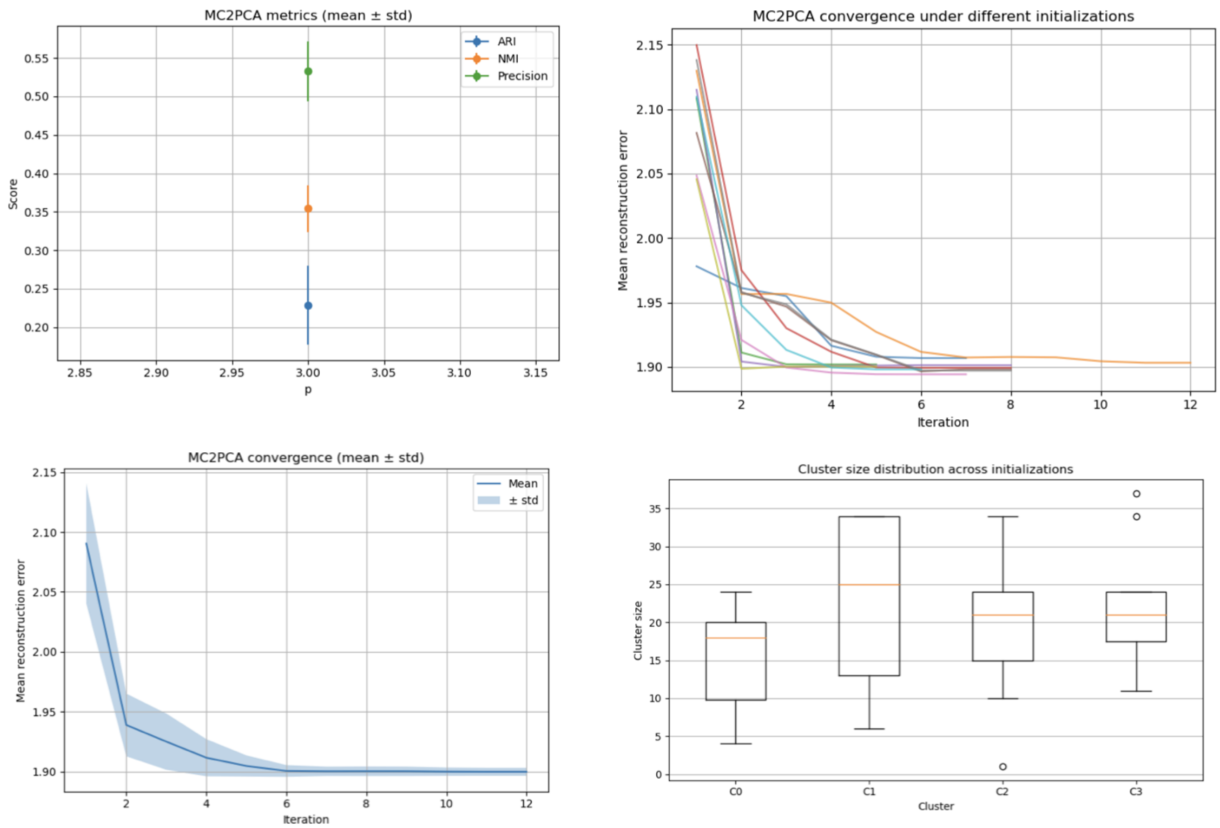

Figure 4: Robustness analysis of MC2PCA on the Basic Motions dataset ($p = 3$). Top left: clustering metrics (ARI, NMI, Precision) averaged over multiple initializations. Top right: convergence curves for different random initializations. Bottom left: mean reconstruction error with standard deviation. Bottom right: cluster size distribution across runs.

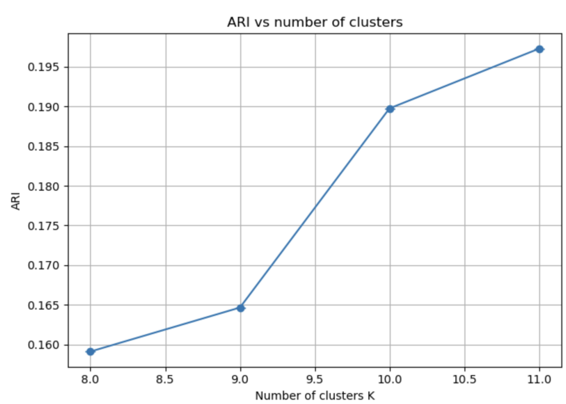

Figure 5: ARI as a function of the number of clusters K on the Spoken Arabic Digits dataset.

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